QUERRIEL ARVY MENDOZA

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EDUCATION

Ph.D. in Computer Science — Kansas State University, Manhattan, KS Expected 2027
GPA 3.9 / 4.0 | Research: Embedded systems, edge AI, and Micro-ROS on NVIDIA Jetson Orin and ESP32

M.S. in Computer Science — Kansas State University, Manhattan, KS 2023
GPA 3.8 / 4.0 | Thesis: Real-time deep learning deployment for insect detection on NVIDIA Jetson Nano

B.S. in Electronics Engineering — Adamson University, Manila, Philippines
GPA 3.75 / 4.0

TECHNICAL SKILLS

Embedded & Robotics: NVIDIA Jetson Orin/Nano, JetPack SDK, ESP32, Micro-ROS, ROS 2, FreeRTOS, DDS / XRCE-DDS, Embedded Linux, I²C, SPI, UART, System Design & Verification (SysML v2, MBSE)
Languages & AI: C, C++, Python, SQL, Rust | PyTorch, TensorFlow, scikit-learn, TensorRT, ONNX, MobileNet-SSD, CNNs, Edge Inference, Computer Vision
Tools & Platforms: Docker, GitHub Actions, MLflow, FastAPI, HL7 FHIR, PySpark, Pandas, Linux, AWS, Azure DevOps

RESEARCH EXPERIENCE

Graduate Research Assistant — Embedded AI, Robotics & Computer Vision Aug 2021 – Present
Kansas State University, Department of Computer Science | Manhattan, KS

- Architecting heterogeneous robotics systems pairing NVIDIA Jetson Orin (perception/learning) with ESP32 (real-time control) via Micro-ROS over DDS-XRCE for edge-AI on resource-constrained hardware.
- Building real-time Micro-ROS publisher / subscriber nodes in C / C++ on ESP32 with FreeRTOS, characterizing end-to-end latency and reliability of sensor streaming to Jetson Orin.
- Built a real-time insect-detection pipeline (85–90% accuracy, 81% F1 across lighting conditions) by training MobileNet-SSD on 1,000 annotated images in PyTorch and deploying via ONNX → TensorRT on Jetson Nano.

PROJECTS

ReadmitRisk — FHIR-Native 30-Day Hospital Readmission Prediction Oct 2026
Personal project | *github.com/qasmendoza/readmit-risk*

- Built an end-to-end system that turns HL7 FHIR R4 records into calibrated 30-day readmission risk scores, served as FHIR RiskAssessment resources by a FastAPI service with a citation-checked RAG care assistant.
- Outperformed the LACE clinical baseline on the latest 1,314 of 6,570 synthetic hospitalizations (AUROC 0.803 vs. 0.752, DeLong $p < 0.001$; calibration slope 1.05) using leakage-safe point-in-time features.Automated the ML lifecycle with MLflow tracking and registry, a quality gate that blocks weaker models, GitHub Actions CI (98 tests, 95% coverage, Docker build), Kubernetes manifests, and PSI + SMD drift monitoring.

PROFESSIONAL EXPERIENCE

Data Science Intern — Caterpillar Inc., Chicago, IL May – Aug 2023

- Generated ~$2 M in projected cost savings by building and deploying ML models in Python / SQL to predict asset and event criticality, surfaced via Power BI for senior stakeholders.
- Delivered insights from large operational datasets with a cross-functional agile team in Azure DevOps.

IT Operations Lead — Accenture, Manila, Philippines Oct 2015 – Oct 2016

- Improved client system performance by 30% (Value Creator Award) with new storage and backup designs.
- Led a 30+ person multi-platform IT operations team that met its SLA and KPI targets.

Infrastructure Engineer (Storage & Backup) — Ingram Micro, Manila, Philippines Jul 2014 – Oct 2015

- Designed and administered storage and backup for new servers on EMC, DataCore, and NetApp systems.

Service Delivery Consultant (Storage & Backup) — HP Enterprise, Manila, Philippines May 2010 – Jul 2014

- Increased system productivity by 25% (Gold Award) with storage and backup designs for new servers.
- Cut the ticket backlog by 45% (High Profile Award) through platform migrations and problem management.

PUBLICATIONS

- **Mendoza, Q. A.**, Chandana, P., & Neilsen, M. (2026). "Predicting the Future of PC Gaming: An Analysis of Genre Popularity, Player Engagement, and Market Trends for 2026-2031." Presented at the *International Conference on Applied Computing*, Las Vegas, NV. In press, Springer Nature.
- Serfa Juan, R. O., **Mendoza, Q. A.**, Neilsen, M., & Gerken, A. (2026). "AI-Driven Morphometric Fragment Analysis for Automated Post-Harvest Identification of Tenebrionidae Species." *IEEE Green Technologies Conf. (GreenTech)*, 470–475. DOI: 10.1109/GreenTech68285.2026.11471609
- **Mendoza, Q. A.**, Pordesimo, L., Neilsen, M., Armstrong, P., Campbell, J., & Mendoza, P. T. (2023). "Application of Machine Learning for Insect Monitoring in Grain Facilities." *AI (MDPI)*, 4(1), 348–360. DOI: 10.3390/ai4010017
- **Mendoza, Q. A.**, et al. (2023). "Enhancing Grain Facility Management with an AI-Based Insect Detection and Identification System." *SPIE Defense + Commercial Sensing*, Paper 12545-33. DOI: 10.1117/12.2672253

CERTIFICATIONS

Machine Learning Specialization (DeepLearning.AI) • DevOps on AWS (AWS) • Data Analysis & Visualization (IBM)